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PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 01 – Lecture 01: AI Overview, Techniques & Applications

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February 28, 2026

Topic focus: **What is AI?** where it is used, and how to judge success

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Rich and Knight, *Artificial Intelligence*, McGraw Hill, 2017.;Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 1–2 (Introduction, Intelligent Agents).;Poole and Mackworth, *Artificial Intelligence: Foundations of Computational Agents* (2e), selected sections.

Quick Links

Overview Applications Models Example Summary

Agenda

Overview

Applications

Models and Success Criteria

Mini Example

Summary

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- Describe what a **model** is and outline **criteria of success** beyond accuracy.

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- **Unit V:** Applications + responsible AI considerations.

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- **KB**: Knowledge Base (used heavily in Unit II)

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Practical takeaway: AI performance depends on **data + compute + modeling choices**, not only on algorithms.

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AI vs ML vs Data Science (Quick Contrast)

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Example

Spam filtering: rule-based filters (AI), learned classifier (ML), reporting spam trends (data science).

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AI Techniques (High-Level Buckets)

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Typical question:

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- **Hybrid**: combine rules + learning (common in real systems)

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Symbolic vs Statistical vs Hybrid (Comparison)

Approach	Strengths	Limitations
Symbolic (rules/logic)	Interpretable; uses explicit knowledge; strong constraints	Hard to scale to messy data; brittle if rules incomplete
Statistical / ML	Learns from data; handles noisy patterns; adapts with retraining	Needs data; can be biased; harder to explain; sensitive to drift
Hybrid	Combines constraints + learning; often best in practice	More engineering; needs careful integration and monitoring

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- **Decision making:** choose an action (agent behavior)

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Key idea: most applications reduce to tasks like **prediction**, **ranking**, **classification**, or **decision making**.

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- **Security** → intrusion detection → anomaly detection + adversaries

Typical AI System Pipeline

Data / Knowledge → Processing → Model → Decision

- In Unit II we focus on **knowledge + logic-based inference**.
- In Units III–IV we focus on **data + learning from examples**.

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- **System level**: code + data pipeline + monitoring in production

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- **Maintainability**: monitoring, retraining, drift handling

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- **Unmonitored drift**: performance degrades over time unnoticed

Worked Example: Spam Filtering (Two Approaches)

Rule-based (symbolic)

IF email contains “win money” AND sender unknown THEN spam.

Learning-based (ML)

Train a classifier on labeled emails using features (word counts, sender reputation, links).

- Rule-based is transparent but brittle.
- ML is adaptable but needs data and careful evaluation.

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This is why "criteria of success" must be tied to real-world costs.

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- So we may prioritize **recall** (sensitivity) over raw accuracy.
- Deployment requires monitoring, calibration, and human oversight.

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Summary

- AI is a broad field: perception + reasoning + learning + action.
- ML is one set of techniques inside AI; data science focuses on insight and value from data.
- Success must be measured with **multiple criteria**, not just accuracy.
- Next: intelligent agents (Unit 01 Lecture 02) and then logic-based AI (Unit II).

Sources: Rich and Knight, *Artificial Intelligence*, McGraw Hill, 2017.;Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 1–2 (Introduction, Intelligent Agents).;Poole and Mackworth, *Artificial Intelligence: Foundations of Computational Agents* (2e), selected sections.

Exit Questions

1. Give two definitions/views of AI (e.g., acting rationally vs thinking humanly).
2. Differentiate AI vs ML vs Data Science with one example each.
3. List 5 criteria of success for an AI system (beyond accuracy).
4. Describe one real-world failure mode caused by data issues.